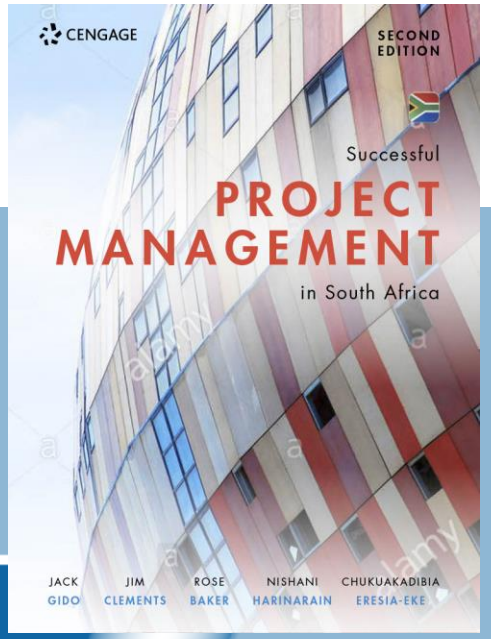
CHAPTER 7



Determining Costs, Budget and Earned Value

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Chapter Concepts

- Estimating the costs of activities
- Determining a time-phased baseline budget
- Determining the earned value of the work performed
- Analysing cost performance
- Forecasting project cost at completion
- Controlling project costs
- Managing cash flow

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Learning Outcomes

- Estimate the cost of activities
- Aggregate the total budgeted cost
- Develop a time-phased baseline budget
- Describe how to accumulate actual costs
- Determine the earned value of work performed
- Calculate and analyse key project performance measures
- Discuss and apply approaches to control the project budget
- Explain the importance of managing cash flow

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Performance Domains of PMBOK® 7th edition covered in this chapter:	Knowledge Areas of PMBOK® 6th edition covered in this chapter:
• Team • Planning • Project Work • Delivery • Measurement • Uncertainty	• Project Integration Management • Project Cost Management
Project Management Principles relevant to this chapter	
Principle 4 (Focus on Value) **Principle 7** (Tailor Based on Context)	

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Estimate Activity Costs

Elements

- Labour
- Materials
- Equipment
- Facilities
- Subcontractors and consultants
- Travel
- Reserve

Good practices

- Have the person responsible estimate costs
- Use historical data to inform current project
- Be reasonable and realistic
- Estimate near-term activities more accurately
- Elaborate other costs as additional information known

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Consumer market study project estimated costs

FIGURE 7.1 Estimated Activity Costs for Consumer Market Study Project

ACTIVITY	NAME	WORK DAYS	LABOUR RATE	LABOUR COSTS	MATERIALS COSTS	TRAVEL COSTS	TOTAL COSTS
1. Identify target consumers	Mbali	3	R260	R780			R780
2. Develop draft questionnaire	Mbali	10	260	2 600			2 600
3. Pilot-test questionnaire	Mbali	20	260	5 200		R3 000	8 200
4. Review comments & finalise questionnaire	Mbali	5	260	1 300			1 300
5. Prepare mailing labels	Fani	2	200	400			400
6. Print questionnaire	Fani	1	200	200	R1 700		1 900
7. Develop data analysis software	S'bu	12	300	3 600			3 600
8. Develop software test data	Mbali	2	260	520			520
9. Mail questionnaire & get responses	Fani	5	200	1 000	7 800		8 800
10. Test software	S'bu	5	300	1 500			1 500
11. Input response data	Jim	7	400	2 800			2 800
12. Analyse results	Jim	8	400	3 200			3 200
13. Prepare report	Jim	10	400	4 000			4 000
		90		**R27 100**	**R9 500**	**R3 000**	**R39 600**

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Determine Project Budget



- The project budgeting process has two steps:

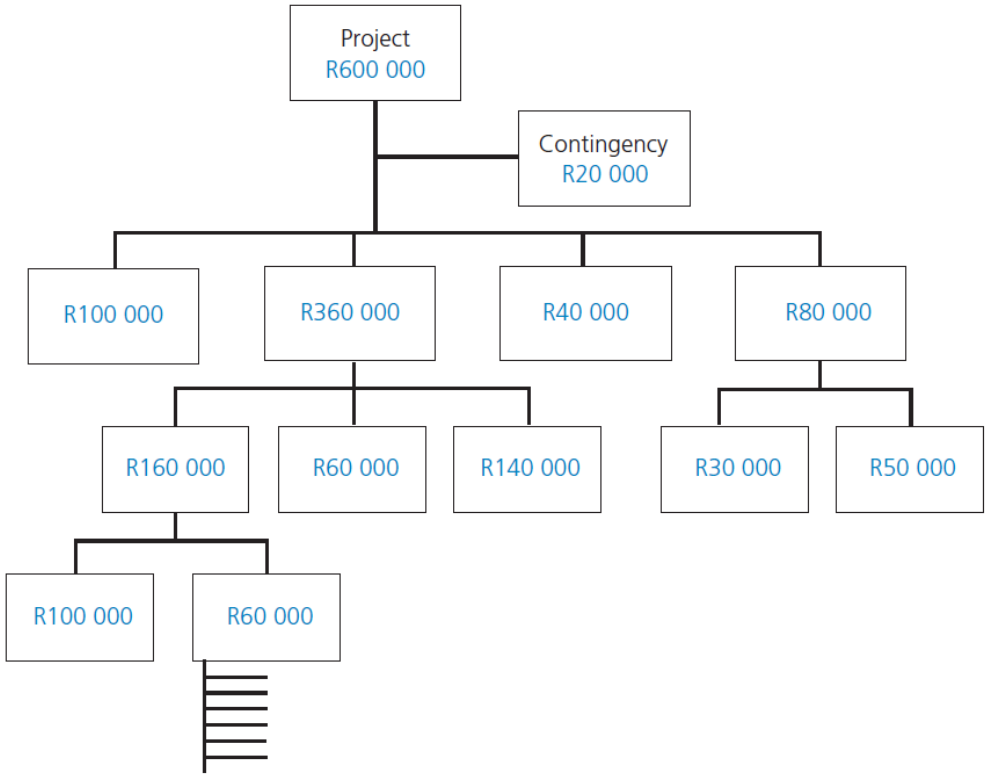
1. First, the project cost estimate is allocated to the various work packages in the project work breakdown structure
2. Second, the budget for each work package is distributed over the duration of the work package

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Aggregate total budgeted cost

- Establish a TBC for each work package
- Determine the process
  - Top-down
  - Bottom-up
- If sum of initial estimates exceeds sponsor budget, then reduce costs and recalculate

FIGURE 7.2 Work Breakdown Structure with Work Package Budgets



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Packaging machine project aggregate total budgeted cost

FIGURE 7.3 Network Diagram for the Packaging Machine Project

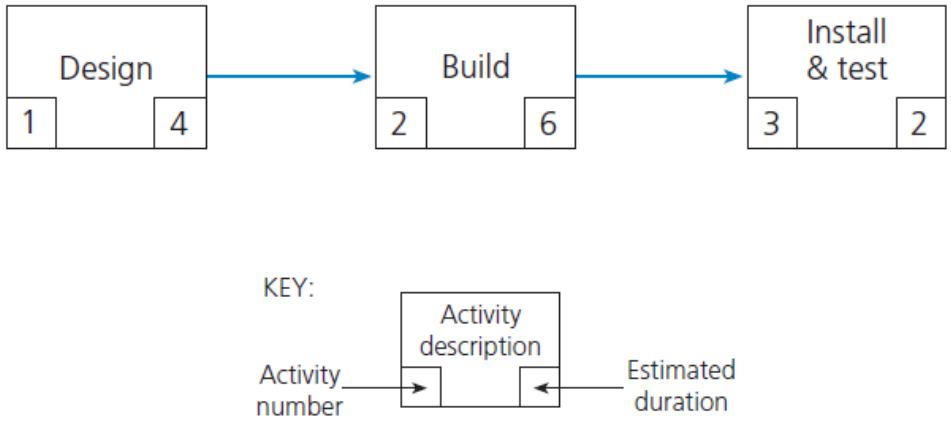
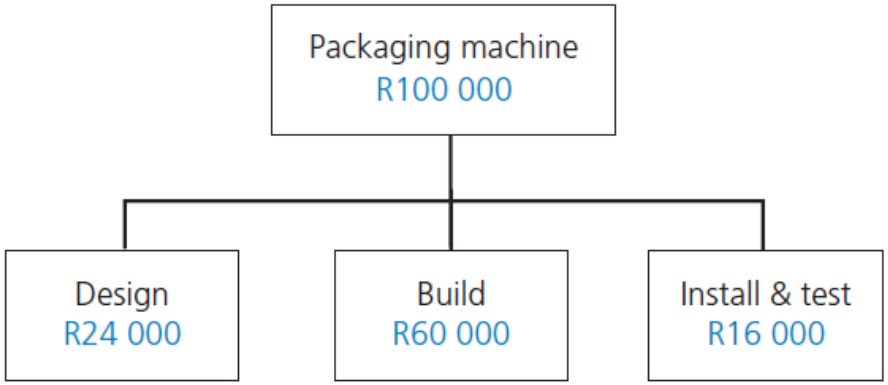


FIGURE 7.4 Work Breakdown Structure for the Packaging Machine Project



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Develop cumulative budgeted cost

- Distribute each total budgeted cost (TBC) over work package duration
- Create the time-phased budget
- Calculate cumulative budgeted cost
- Provides a baseline against which actual cost and work performance are measured

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Determine Actual Cost

- Actual cost
  - Collect data regularly for funds actually expended
  - Charge to work package numbers
- Committed costs
  - Periodically assign portion of total cost to actual cost
  - Include costs for items that will be paid for later
- Compare actual cost to budgeted cost
  - Calculate cumulative actual cost
  - Compare to cumulative budgeted cost

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Packaging machine project determine actual cost

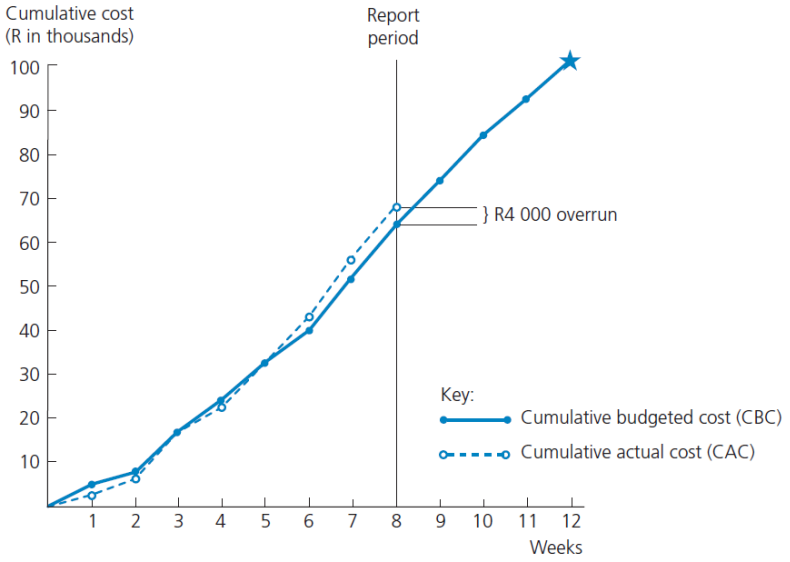
- End of Week 8
  - Planned cost = R64,000
  - Actual cost = R68,000
- Compare CAC with CBC

FIGURE 7.7 Actual Cost by Period for the Packaging Machine Project

	Week								Total expended
	1	2	3	4	5	6	7	8	
Design	2	5	9	5	1				22
Build				2	8	10	14	12	46
Install & test									0
Total	2	5	9	7	9	10	14	12	**68**
Cumulative	**2**	**7**	**16**	**23**	**32**	**42**	**56**	**68**	**68**

Amounts are in thousands of rands.

FIGURE 7.8 Cumulative Budgeted and Actual Costs for the Packaging Machine Project



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Determine Value of Work Performed

Example project

- Paint 10 similar rooms
- Total budgeted cost of R2 000
- Budget is R200 per room

At Day 5

- R1 000 has been spent
- 3 rooms have been painted
- Earned value =

  0.30 X R2 000 = R600
- Have expended R400 more than the earned value

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Analyse Cost Performance

- Four cost-related measures
  - TBC – total budgeted cost
  - CBC – cumulative budgeted cost
  - CAC – cumulative actual cost
  - CEV – cumulative earned value
- Use to analyse project cost performance
- Plot CBC, CAC, and CEV curves on the same graph
  - Reveal any trends toward improving or deteriorating cost performance

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Cost performance index

- Measure of the cost efficiency with which the project is being performed

- Cost performance index =

  Cumulative earned value/Cumulative actual cost

  CPI = CEV/CAC

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Packaging machine project cost performance index

End of Week 8

- R64 000 was budgeted
- R68 000 was actually expended
- R54 000 was the earned value of work actually performed

- CEV = R54 000
- CAC = R68 000

Determine CPI

- CPI = CEV/CAC
  = R54 000/R68 000
  = 0.79

  For every R1.00 actually expended, only R0.79 of earned value was received.

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Cost variance

- Indicator of cost performance
- Difference between the cumulative earned value of the work performed and the cumulative actual cost

- Cost variance =

  Cumulative earned value – Cumulative actual cost

$$CV = CEV - CAC$$

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Packaging machine project cost variance

End of Week 8

- R64 000 was budgeted
- R68 000 was actually expended
- R54 000 was the earned value of work actually performed

- CEV = R54 000
- CAC = R68 000

Determine CV

- CV = CEV – CAC

  = R54 000 – R68 000

  = –R14 000

  The value of the work performed through week 8 is R14 000 less than the amount actually expended.

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Estimate Cost at Completion

- Forecast what the total costs will be at the completion of the project or work package
- 3 different methods
  - FCAC = TBC/CPI
  - FCAC = CAC + (TBC – CEV)
  - FCAC = CAC + Re-estimate of remaining work
- Another method
  - TCPI = (TBC – CEV)/(TBC – CAC)

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Packaging machine project estimate cost at completion

End of Week 8

- R64 000 was budgeted
- R68 000 was actually expended
- R54 000 was the earned value of work actually performed

- CEV = R54 000
- CAC = R68 000
- CPI = 0.79
- TBC = R100 000

Determine FCAC

- FCAC = TBC / CPI
  = R100 000/0.79 = R126 582

- FCAC = CAC + (TBC – CEV)
  = R68 000 + (R100 000 – R54 000)
  = R68 000 + R46 000
  = R114 000

- TCPI = (TBC – CEV)/(TBC – CAC)

= (R100 000 − R54 000)/(R100 000 − R68 000)
= R46 000/R32 000
= 1.44

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Important Project Costs, Budget and Earned Value Concepts

- TBC

Total Budgeted Cost - The sum of the estimated costs of all the specific activities that make up that work package or the project

- BAC

Budget at completion (same as TBC)

- CBC

Cumulative Budgeted Cost – The amount that is budgeted to complete all the work that is scheduled to be performed up to a specific point in time.

FIGURE 7.5 Budgeted Cost by Period for the Packaging Machine Project

		Week											
	TBC	1	2	3	4	5	6	7	8	9	10	11	12
Design	24	4	4	8	8								
Build	60					8	8	12	12	10	10		
Install & test	16											8	8
Total	100	4	4	8	8	8	8	12	12	10	10	8	8
Cumulative		**4**	**8**	**16**	**24**	**32**	**40**	**52**	**64**	**74**	**84**	**92**	**100**

Amounts are in thousands of rands.

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- PV

Planned value (same as CBC)

- BCWS

Budgeted cost of work scheduled (same as CBC)

- CAC

Cumulative actual cost – The amount of costs actually expended and committed to complete all the work performed up to a specific point in time

FIGURE 7.5 Budgeted Cost by Period for the Packaging Machine Project

		Week											
	TBC	1	2	3	4	5	6	7	8	9	10	11	12
Design	24	4	4	8	8								
Build	60					8	8	12	12	10	10		
Install & test	16											8	8
Total	100	4	4	8	8	8	8	12	12	10	10	8	8
Cumulative		**4**	**8**	**16**	**24**	**32**	**40**	**52**	**64**	**74**	**84**	**92**	**100**

Amounts are in thousands of rands.

FIGURE 7.7 Actual Cost by Period for the Packaging Machine Project

	Week								Total expended
	1	2	3	4	5	6	7	8	
Design	2	5	9	5	1				22
Build				2	8	10	14	12	46
Install & test									0
Total	2	5	9	7	9	10	14	12	**68**
Cumulative	**2**	**7**	**16**	**23**	**32**	**42**	**56**	**68**	**68**

Amounts are in thousands of rands.

- ACWP

Actual cost of work performed (same as CAC)

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- EV

Earned Value – The value of work actually performed – involves collecting data on the % complete for each work package and then converting this % to a rand amount by multiplying the TBC of the work package by the % complete

- BCWP

Budgeted cost of work performed (same as EV)

- CEV

Cumulative earned value – the value of the work actually performed up to a specific point in time.

Calculated by multiplying the TBC by the % of the work estimated to be completed.

Example: painting 10 rooms cost R2000, if 3 rooms were completed, we can say 30% of the work has been performed. The CEV is – 0.30 X R2000 = R600

- CPI

Cost performance index – a measure of the cost efficiency with which the project is being performed.

The formula for determining the CPI is:

$$\textbf{CPI} = \frac{\textit{Cumulative earned value}}{\textit{Cumulative actual cost}} = \frac{CEV}{CAC}$$

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- CV

Cost Variance – the difference between the CEV of the work performed and the CAC

The formula for determining the CV is:

CV = CEV - CAC

- FCAC

Forecasted cost at completion – referred to as the Estimated cost at completion (EAC)

What the total costs will be at completion of the project or work package based on the actual cost expended, the earned value of the work performed and the remaining work to be done.

The are 3 methods to calculating FCAC:

Method 1 - $\mathbf{FCAC} = \frac{\boldsymbol{TBC}}{\boldsymbol{CPI}}$

Method 2 - $\mathbf{FCAC} = \mathbf{CAC} + (\mathbf{TBC} - \mathbf{CEV})$

Method 3 - $\mathbf{FCAC} = \mathbf{CAC} + \mathbf{Re} - \mathbf{estimate\ of\ the\ remaining\ work\ to\ be\ performed}$

- TCPI

To-complete performance index – it is the rate of performance required for the remaining work to complete the project or work package within its TBC

The formula for TCPI is:

$$\mathbf{TCPI} = \frac{(\boldsymbol{TBC} - \boldsymbol{CEV})}{(\boldsymbol{TBC} - \boldsymbol{CAC})}$$

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Control Costs

- Analyse cost performance on a regular basis
  - Determine which work packages require corrective action
  - Decide what specific corrective action
  - Revise the project plan
- Evaluate negative cost variance
- Take corrective actions
  - Near term activities
  - Activities with large cost estimate
- Reduce costs of activities
- Evaluate the trade-off of cost and scope

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- What is the **key** to managing cash flow?
- How would **cash flow** on a project be **managed**?
- How would **cash flow** on a project be **controlled**?

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Manage Cash Flow

- Ensure that cash comes in faster than it goes out
- Negotiate payment terms
  - Provide a down payment
  - Make equal monthly payments
  - Provide frequent payments
- Avoid only one payment at end of project
- Control outflow of cash

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Cost Estimating for Information Systems Development

- Common errors in estimating costs
  - Underestimating the work time necessary to complete an activity
  - Requiring rework to meet the user requirements
  - Underestimating growth in the project scope
  - Not anticipating new hardware purchases
  - Making corrections to flaws in excess of the reserve planning
  - Changing the design strategy
  - Increasing resources to fast-track phases of the SDLC

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Project Management Information Systems

- Store all costs associated with each resource
- Calculate the budget for each work package
- Determine cost for the entire projects
- Define different rate structures for each resource
- Analyse cost performance

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Critical Success Factors

- Estimated activity **costs** must be **based on** the estimated **activity resources**.
- The **person** who will be **responsible** for performing the activity should **estimate the costs** for that activity. This generates commitment from the person.
- Cost estimates should be **reasonable and realistic**.
- Once the project starts, it is important to **monitor actual costs and work performance** to ensure that everything is within budget.
- A system should be established to **collect**, on a regular and timely basis, data on **costs actually expended and committed**, and the earned value (percent complete) of the work performed, so they can be compared to the cumulative budgeted cost (CBC).
- If at any time during the project it is determined that the project is overrunning the budget, or the value of the work performed is not keeping up with the actual amount of costs expended, **corrective action must be taken immediately**.

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Critical Success Factors (continued)

- It is important to use the **time-phased cumulative budgeted cost** (CBC), rather than the total budgeted cost (TBC), as the **baseline** against which cumulative actual cost (CAC) is compared. It would be misleading to compare the actual costs expended to the total budgeted cost because cost performance will always look good as long as actual costs are below the TBC.
- To permit a realistic comparison of cumulative actual cost to cumulative budgeted cost, portions of the **committed costs** should be **assigned** to **actual costs** while the associated work is in progress.
- The **earned value** of the work actually performed is a key parameter that must be determined and reported throughout the project.
- For each reporting period, the **percent complete data** should be obtained from the person responsible for the work. It is important that the person make an honest assessment of the work performed relative to the entire work scope.
- One way to prevent inflated percent complete estimates is to keep the work packages or activities small in terms of scope and duration. It is important that the person estimating the percent complete assess not only how much work has been performed but also what **work remains to be done**.

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Critical Success Factors (continued)

- The key to effective cost control is to **analyse cost performance** on a **timely and regular basis**. Early identification of cost variances (CV) allows corrective actions to be taken immediately, before the situation gets worse.
- For analysing cost performance, it is important that all the **data** collected be as **current as possible** and be based on the same reporting period.
- Trends in the **cost performance index** (CPI) should be monitored carefully. If the CPI goes **below 1.0** or gradually decreases, **corrective action** should be taken.
- As part of the regular cost performance analysis, the estimated or **forecasted cost at completion** (FCAC) should be **calculated**.
- The key to effective cost control is to **aggressively address work packages** or activities **with negative cost variances** and cost inefficiencies as soon as they are identified. A concentrated effort must be applied to these areas. The amount of negative cost variance should determine the priority for applying these concentrated efforts.

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Critical Success Factors (continued)

- When attempting to reduce negative cost variances, **focus on** activities that will be performed in the **near term** and on activities that have **large estimated costs**.
- Addressing cost problems early will **minimise the negative impact** on scope and schedule. Once costs get out of control, getting back within budget becomes more difficult and is likely to require reducing the project scope or quality, or extending the project schedule.
- The key to **managing cash flow** is to ensure that cash comes in faster than it goes out.
- It is desirable to **receive payments** (cash inflow) from the customer as **early as possible**, and to delay making payments (cash outflow) to suppliers or subcontractors as long as possible.

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Summary

- The total project cost is often estimated during the initiating phase of the project when the project charter or a proposal is prepared, but detailed plans are not usually prepared at that time.
- The project budgeting process involves two steps: the budget for each work package is determined and the budget for each work package is then distributed over the expected time.
- Aggregating the estimated costs of the specific activities for the appropriate work packages in the work breakdown structure will establish a total budgeted cost (TBC).
- The cumulative budgeted cost (CBC) is the time-phased baseline budget that will be used to analyse the cost performance of the project.
- At any time during the project, it is possible to forecast what the total costs will be at the completion of the project or work package based on analysis of actual cost expended and the earned value of work performed.
- The key to effective cost control is to analyse cost performance on a regular and timely basis.
- It is important to manage the cash flow on a project.

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